

SYED SURKHI

ENGINEERING/SOFTWARE COOP

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RE: Application for Embedded Developer Intern, Video Products (Summer 2026)

Dear Geotab Hiring Team,

I am a Computer Engineering co-op student at Toronto Metropolitan University applying for the Embedded Developer Intern, Video Products role for Summer 2026. My background in C, C++, Python, and embedded Linux, along with hands-on work on Raspberry Pi and microcontrollers, aligns with your focus on firmware test automation, device data analysis, and system-level validation.

At Toronto MetRobotics, I built a real-time control system for a rover on Raspberry Pi and Arduino. I integrated sensors and actuators over I2C, PWM, GPIO, and serial links, and wrote multithreaded Python to manage inputs, telemetry, and motor control. I also created a multi-camera debugging GUI using PyGST and TCP streaming that improved test visibility and reduced field issues. This work gave me practical experience with device bring-up, Linux tooling, and writing tools that operators can trust.

With Metropolitan Hyperloop, I developed a C++ and Python telemetry GUI using TCP sockets on Linux and set up a Raspberry Pi sensor acquisition stack. I wrote drivers and test scripts, captured logs, and documented procedures so others could reproduce results. That experience maps well to your test automation framework, unit and integration testing, and close work with development and test teams.

I am comfortable with SQL and data workflows. In recent projects I designed data pipelines and wrote queries against PostgreSQL to measure feature health and track regressions. I can bring the same approach to BigQuery and Python scripts for device health dashboards. I keep code clean, write clear commit messages, and use Git every day. I follow coding standards and add focused tests to protect core paths.

Beyond embedded work, I have shipped production services in Python and FastAPI, set up CI/CD, and reviewed code in team settings. I write concise documentation and share progress in simple terms. I learn new tools quickly and enjoy digging into device behavior until I can explain it with data.

I am available full-time from May 2026 for a 4 to 12 month term and can work from Oakville or Waterloo. I would value the chance to deepen my embedded skills on GO and IOX devices, extend your Python and C test frameworks, and help build dashboards that make device behavior clear.

Thank you for your time and consideration. I would welcome the opportunity to discuss how I can contribute to Geotab's video products team.

Sincerely,
Syed Kaab Surkhi